

Enhanced Speech Processing with Learnable Verbalizers for Accurate Classification

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1 Abstract

This project introduces SpeechPrompt, a groundbreaking framework that leverages the prompting paradigm for Statistical Language Model (SLM) to address a wide array of speech processing tasks efficiently and effectively. Traditional approaches, such as the "pre-train, fine-tune" paradigm, demand task-specific model designs, significant computational resources, and extensive parameter tuning, which limit scalability and increase storage requirements. SpeechPrompt overcomes these challenges by reformulating speech processing tasks—such as speech classification, sequence generation, and speech synthesis—into a unified speech-to-unit generation framework. It utilizes discrete speech units derived from quantized Self Supervised Learning (SSL) representations, which encode essential phonetic, acoustic, and linguistic information while being computationally efficient and storage-friendly. These discrete units are further enhanced through a learnable verbalizer, bridging them with downstream labels to improve task performance, explainability, and adaptability. Experimental evaluations reveal that SpeechPrompt achieves competitive results against strong fine-tuning baselines, with particular strengths in low-resource and few-shot learning settings, thereby highlighting its ability to generalize with minimal training. Additionally, the framework demonstrates scalability for cloud-based deployment and the capacity to process diverse languages, including those without standard textual representations. The study emphasizes the transformative potential of prompting SLM, paving the way for future innovations like zero-shot learning and cross-modal integration while fostering advancements in universal and efficient speech processing technologies.

2 Introduction

Speech processing has undergone significant advancements in recent years, driven by the integration of machine learning techniques, especially SSL. These advancements have made it possible to extract meaningful representations from speech without requiring large labeled datasets, which have traditionally been a barrier to the scalability of speech technologies. In particular, SSL models like Wav2Vec2 have demonstrated exceptional performance in learning robust feature representations from raw audio, without the need for manual feature engineering. This breakthrough has opened up new possibilities for diverse speech applications, from speech recognition to sentiment analysis and beyond.

Traditional speech processing methods often rely on task-specific models, which are typically designed to solve one problem at a time (e.g., Automatic Speech Recognition (ASR) or speaker identification). These models require extensive training data for each task, which can lead to inefficiencies and limited scalability. Moreover, they often involve significant fine-tuning of pre-trained models, which can increase the complexity and computational cost of deploying speech technologies. The challenge of creating a more scalable and generalizable approach for speech processing has led to the development of frameworks that leverage SSL and prompting methodologies, which can be applied to multiple speech tasks without needing to retrain models for each specific task.

One of the most promising methodologies in modern speech processing is the SpeechPrompt framework, which reformulates speech tasks into a uniform task definition. By leveraging a prompting paradigm, SpeechPrompt enables the application of a single pre-trained language model to a variety of speech processing tasks. The concept of prompting, commonly used in Natural Language Processing (NLP), allows for task-specific behavior to be learned through the use of carefully designed prompts, without needing to modify the underlying model architecture. This approach not only reduces the need for task-specific models but also improves computational efficiency by eliminating the need to fine-tune large models for each task [1].

A key component of this framework is the use of SSL models, such as Wav2Vec2, which convert raw speech into high-dimensional feature representations that can be used for downstream tasks like classification or generation. Wav2Vec2, proposed by Baevski et al., is a self-supervised model that pre-trains on large amounts of unlabeled speech data, learning to capture various acoustic features of speech. Once trained, Wav2Vec2 can be fine-tuned for specific tasks, such as ASR, by simply adapting its representations using task-specific data. This approach has drastically improved the performance of ASR systems, especially in low-resource environments, by reducing the need for large labeled datasets [2].

The traditional method of speech processing typically involves extracting features from speech signals and then using supervised learning algorithms to classify or model the data. These methods often require domain-specific expertise for feature extraction, making them less flexible and scalable for different applications. In contrast, the speech-to-unit generation paradigm, which underlies this project, allows for the automatic quantization of speech features into discrete units. These units can then serve as the foundation for a variety of tasks, including speech synthesis, sequence generation, and classification. The use of k-means clustering to quantize speech features into discrete units has been shown to be an effective approach for improving the efficiency and generalizability of speech processing systems [3].

In addition to model performance, the computational efficiency of speech processing systems is a critical consideration, especially for real-world applications that require real-time processing or deployment on resource-constrained devices. Traditional methods of speech processing, including those based on deep learning, often require significant computational resources, including high-performance Graphics Processing Unit (GPU)s or cloud-based infrastructure. The use of task-agnostic discrete units allows for more efficient storage and computation by reducing the complexity of the speech representations that need to be processed. Furthermore, by using a single framework for multiple tasks, the system can be easily scaled to accommodate new applications or languages with minimal additional computational overhead [7].

The goal of this project is to implement a unified framework for speech processing that leverages SSL models and discrete unit generation to tackle various tasks in a computationally efficient manner. By reformulating speech tasks into a speech-to-unit generation paradigm, this approach simplifies the integration of new speech-related applications, such as speaker verification, speech-to-text conversion, and speech synthesis, without requiring extensive re-engineering of the underlying models. This is particularly beneficial for organizations or researchers working in low-resource or multilingual environments, where large labeled datasets may not be readily available.

This project will focus on implementing a modular and interactive framework using an Interactive Python Notebook (IPYNB) format, which allows users to experiment with the system, evaluate its performance across different tasks, and visualize the results. The framework will be designed with flexibility in mind, allowing for easy adaptation to future developments in speech processing. Evaluation and benchmarking will be carried out to compare the performance of this unified approach with that of traditional task-specific models, particularly in terms of classification accuracy, Word Error Rate (WER), and processing speed.

The impact of this work extends beyond improving existing speech processing systems; it also lays the foundation for future innovations in the field. By offering a scalable, efficient, and adaptable solution to speech processing, this project paves the way for further research into low-resource speech technologies and the development of applications that can function across different languages, dialects, and environments. As the field of speech processing continues to evolve, the integration of SSL models, prompting paradigms, and discrete unit generation will undoubtedly play a pivotal role in shaping the next generation of speech technologies.

3 Motivation & Background

Prompting has emerged as a powerful method for leveraging pre-trained Language Model (LM) in solving a diverse range of tasks with minimal computational and storage requirements. By modifying only the inputs to the model, the prompting paradigm has demonstrated its ability to unify downstream tasks, eliminating the need for task-specific models and fine-tuning. This approach has gained considerable attention in the NLP domain and is now being explored for speech processing.

The research paper, [1], serves as a key motivation for this project. It introduces the concept of reformulating speech processing tasks into speech-to-unit generation tasks using discrete speech units. These units are derived from self-supervised learning (SSL) representations and quantized through methods like k-means clustering. The SpeechPrompt framework achieves competitive performance across tasks such as speech classification, sequence generation, and speech synthesis, highlighting its potential to simplify and optimize speech processing workflows.

By addressing limitations of the traditional "pre-train, fine-tune" paradigm, such as computational inefficiency and high storage demands, the SpeechPrompt methodology demonstrates how the prompting paradigm can achieve scalability, adaptability, and efficiency in handling diverse speech-related tasks. Additionally, the use of discrete units offers advantages like reduced storage requirements and enhanced privacy, making it especially valuable in scenarios involving limited resources or sensitive data.

Building on this motivation, our project aims to implement a similar unified framework that leverages the strengths of prompting and speech-to-unit generation. Inspired by the advancements presented in SpeechPrompt, we seek to develop a scalable, efficient, and adaptable solution for modern speech processing challenges.

4 Project Objectives

The primary objective of this project is to design and implement a unified, efficient, and adaptable framework for speech processing tasks using a prompting-based paradigm. By leveraging state-of-the-art self-supervised learning models and a speech-to-unit generation methodology, this project aims to address key challenges in scalability, computational efficiency, and task adaptability. The specific objectives include:

- **Speech Encoding Module Development:** Implement Wav2Vec2 for encoding raw speech into high-dimensional feature representations that capture meaningful speech patterns.
- **Discrete Unit Generation:** Utilize k-means clustering to quantize the extracted speech features into discrete units, enabling flexible task integration.
- **Task-Specific Prompting Framework:** Design and develop task-specific prompts for applications such as speech classification, sequence generation, and speech synthesis, eliminating the need for modifications to the underlying language model.
- **Optimization for Computational Efficiency:** Ensure the framework’s design is computationally efficient, enabling scalability for real-world applications while maintaining high performance.
- **Unified Speech-to-Unit Generation Paradigm:** Reformulate diverse speech tasks into a single speech-to-unit generation paradigm to enhance consistency, reduce complexity, and support future speech technologies.
- **Evaluation and Benchmarking:** Rigorously evaluate the framework’s performance using relevant metrics for each task and compare results against existing state-of-the-art methods to validate efficiency and adaptability.
- **Interactive IPython Notebook Integration:** Deliver the project as a modular and IPYNB, ensuring usability, reproducibility, and accessibility for further research and practical applications.

5 Proposed Methodology

This methodology combines advanced techniques from both clustering and embedding learning with the SpeechPrompt(2024) paper [7] to process, classify, and evaluate speech data. The steps outlined below detail the process from data preprocessing and feature extraction to model training, evaluation, and deployment.

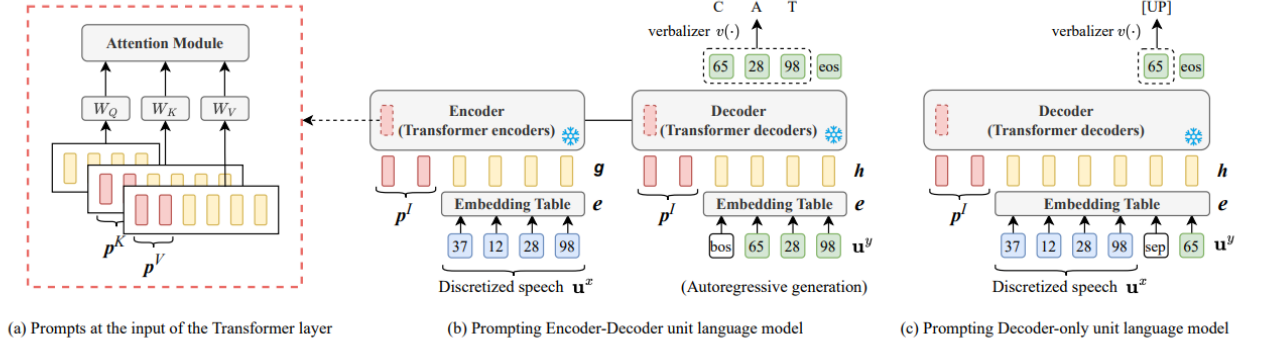


Figure 1: An overview of the proposed framework, where all downstream tasks are treated as speech-to-unit generation processes. The generation of units is directed by the task-specific prompts that guide the unit language model. A verbalizer or speech decoder then bridges the gap between the generated units and the corresponding downstream labels [7].

1. Configuration and Setup

Objective: Set up the environment and initialize necessary libraries and models.

- **Import Libraries:** Load essential libraries, including PyTorch for deep learning, scikit-learn for data processing, and speech-processing libraries like `Wav2Vec2Processor` for feature extraction, `KMeans` for clustering, `Principal Component Analysis` (PCA) for dimensionality reduction, and `LabelEncoder` for class mapping.
- **Pre-trained Model Initialization:** Load pre-trained models such as `Wav2Vec2` for speech feature extraction, setting up a suitable environment for training.
- **Verbalizer Setup:** Initialize a learnable `verbalizer`, which bridges the model’s continuous representations to discrete labels (such as client IDs).

2. Dataset Loading and Preprocessing

Objective: Preprocess speech data and extract meaningful features for clustering and classification.

- **Dataset Loading:** Load the audio dataset (e.g., Waveform Audio File Format (WAV) files) and convert them into a suitable format for processing.

- **Feature Extraction:** Use `Wav2Vec2Processor` to extract features from the speech data. This involves converting raw audio to embeddings or other representations.
 - In the decoder-only LM, the model consists solely of the decoder D without an encoder. The discretized source speech u_x is fed into the decoder at the beginning of the sequence as the initial context, allowing the model to predict subsequent units. A separation token $\langle \text{sep} \rangle$ is inserted between the source speech sequence u_x and the generated units u_y . At each timestep t in the autoregressive process, the input to the decoder D is:

$$h^{(1)} = [e(u_x), e(\langle \text{sep} \rangle), e(u_{y_1}), \dots, e(u_{y_t})]$$

- **Dimensionality Reduction:** Apply PCA to reduce the dimensionality of the extracted features, retaining the most significant information while improving computational efficiency.
- **Clustering with KMeans:** Use `KMeans` to group the extracted features into clusters, allowing for the generation of discrete units that can be used by the verbalizer.

3. Data Splitting

Objective: Split the dataset into training and validation sets for model evaluation.

- **Stratified Sampling:** Perform stratified sampling to ensure proportional representation of each class (e.g., client IDs) in both training and validation sets.
- **Test Set Size Adjustment:** Adjust the test size to maintain a balance between the size of the training and validation sets, addressing class imbalance in speech datasets.

4. Verbalizer Initialization

Objective: Initialize and train the verbalizer to map continuous embeddings to discrete class labels.

- **Verbalizer Setup:** Initialize a learnable verbalizer model that maps continuous feature embeddings to predefined class labels (such as client IDs).
- **Training the Verbalizer:** Use gradient-based optimization to update the verbalizer during training, enabling it to make accurate class predictions based on the embeddings from speech features.
- **The mappings are determined by a learnable linear transformation matrix $W \in \mathbb{R}^{|Y| \times |V|}$, where $|V|$ is the size of the original LM’s vocabulary and $|Y|$ is the number of classes in the downstream task.**
- **Logits Transformation:** The matrix W is applied to the logits vector $z_t \in \mathbb{R}^{|V| \times 1}$ to produce a transformed logits vector $\hat{z}_t \in \mathbb{R}^{|Y| \times 1}$ for the downstream task labels:

$$\hat{z}_t = W \cdot z_t$$

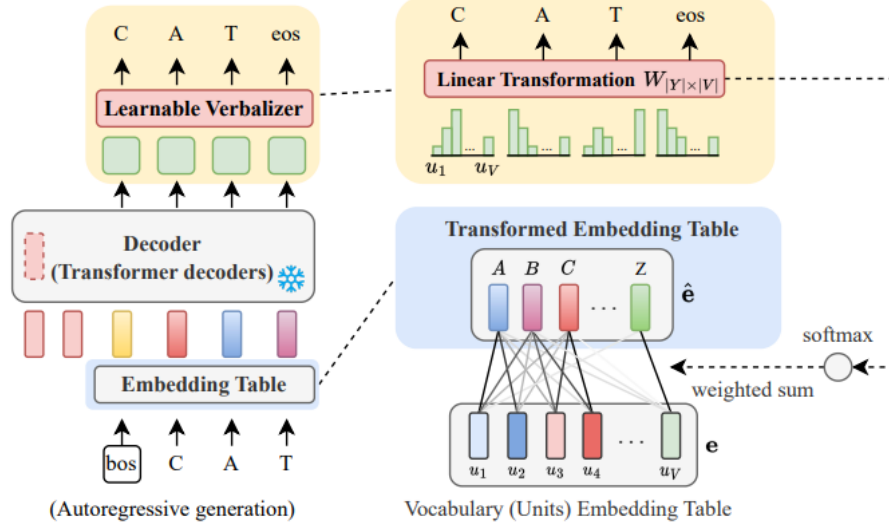


Figure 2: Illustration of the learnable verbalizer. The logits are transformed into labels for the downstream task through a linear transformation. Furthermore, the original vocabulary embeddings are converted into class-specific embeddings using weighted transformations, aligning them more closely with the downstream task [7].

- **Label Prediction:** The label y_t is sampled from the transformed logits:

$$y_t = \arg \max_y P(y|\hat{z}_t)$$

where $P(y|\hat{z}_t)$ is the probability of class y given the transformed logits $\hat{z}_t$ at timestep t .

- **Input Transformation Mechanism:** A mechanism is proposed to transform the original vocabulary embeddings into new embeddings suitable for the downstream task’s labels, coupled with the learnable verbalizer matrix W .
- **Transformed Embedding:** The transformed embedding for a given class y , denoted as $\hat{e}(y)$, is computed as a weighted sum of the original vocabulary embeddings e . The weight vector W_y for class y is processed as:

$$\hat{e}(y) = \sum_{i=1}^{|V|} \text{softmax} \left(\frac{W_y}{\tau} \right)_i \cdot e(u_i)$$

where τ is the temperature parameter and $e(u_i)$ represents the original embeddings.

- **Class-Specific Embeddings:** The resulting $\hat{e}(y)$ is a class-specific embedding tailored to the downstream task, improving the model’s adaptability to varied speech processing applications.

5. Model Training

Objective: Train the model using the extracted features and the verbalizer.

- **Loss Function:** Use **cross-entropy loss** to compare the predicted class labels with the true labels, ensuring that the model learns to classify speech data correctly.
- **Optimization:** Employ the **Adam optimizer** to minimize the loss function and update the verbalizer parameters during training.
- **Epoch Tracking:** Track the average training loss for each epoch to monitor the model's performance over time.

6. Validation and Hyperparameter Tuning

Objective: Evaluate the model on the validation set and adjust hyperparameters to improve performance.

- **Validation Set Evaluation:** Periodically evaluate the model on the validation set to monitor overfitting and ensure the model generalizes well.
- **Hyperparameter Tuning:** Adjust hyperparameters such as learning rate, batch size, or the number of epochs based on the validation performance.
- **Early Stopping:** Implement early stopping to prevent overfitting and ensure the model achieves optimal generalization.

7. Model Evaluation on Test Set

Objective: Evaluate the trained model's performance on a separate test dataset.

- **Generating Discrete Units:** Use the trained model to generate discrete units from test audio samples based on the clustering assignments.
- **Prediction:** Use the verbalizer to predict the class labels (e.g., client IDs) based on the embeddings from the test data.
- **Accuracy Calculation:** Compute accuracy by comparing predicted labels with the true class labels, and evaluate how well the model performs.

8. Enhanced Evaluation Metrics

Objective: Go beyond accuracy by evaluating precision, recall, and F1-score.

- **Precision, Recall, F1-Score:** Compute these additional metrics for each class to gain deeper insights into the model's performance, especially in imbalanced datasets.
- **Class-wise Metrics:** Analyze the performance of the model for each individual class to identify areas for improvement.

9. Confusion Matrix Visualization

Objective: Visualize the distribution of correct and incorrect predictions across classes.

- **Generate Confusion Matrix:** Compute and display the confusion matrix, which shows how predictions are distributed across different classes.
- **Normalization:** Normalize the confusion matrix by true labels to highlight the proportion of correct predictions for each class, providing a clearer analysis of model performance.

10. Display Classification Report

Objective: Provide a detailed report of classification metrics for model assessment.

- **Classification Report:** Generate a detailed classification report that includes precision, recall, F1-score, and support for each class.
- **Model Insights:** Use the report to gain insights into the model's strengths and weaknesses, guiding further refinement.

11. Interactive 3D Visualization of Embeddings

Objective: Visualize the learned embeddings interactively to assess their distribution and relationships.

- **Embedding Visualization:** Use t-SNE or PCA to reduce the dimensionality of the learned embeddings and visualize them in 2D or 3D space.
- **Interactive Plot:** Utilize Plotly or another 3D visualization tool to create interactive plots, allowing for rotation and zoom to explore the embedding space.
- **Embedding Index Hovering:** Show embedding indices or cluster assignments when hovering over the points in the plot to track specific embeddings.

12. Model Saving and Deployment

Objective: Save the trained model for future inference or further fine-tuning.

- **Model Saving:** Save the trained verbalizer model and associated parameters to disk.
- **Deployment:** Use the saved model for inference on new, unseen data or deploy it to production environments for real-world applications.

13. Final Model Evaluation

Objective: Perform a comprehensive evaluation using various metrics to assess the model's overall performance.

- **Cluster Evaluation:** Evaluate the clustering quality using metrics like `silhouette score` or `Davies-Bouldin index` to assess how well the KMeans algorithm has grouped similar features.

- **Cluster and Embedding Interpretation:** Interpret the clustering results by analyzing the distribution of audio segments across clusters and the corresponding embeddings.
- **Model Refinement:** Use insights from the confusion matrix, classification report, and evaluation metrics to refine the model, optimize clustering and verbalizer settings, and improve future iterations.

Interactive IPython Notebook: The implementation will be delivered in an IPYNB format to allow for easy experimentation and reproducibility. The notebook will include modular code blocks and visualizations for the user to interact with the model and evaluate its performance.

Hence this methodology integrates the key techniques from the **SpeechPrompt** paper (2024) with advanced clustering, dimensionality reduction, and embedding learning techniques. By using **Wav2Vec2** for feature extraction, **KMeans** for clustering, **PCA** for dimensionality reduction, and a **learnable verbalizer** for mapping continuous representations to class labels, the model can efficiently process and classify speech data. The integration of detailed evaluation metrics, confusion matrices, and interactive visualizations ensures that the model’s performance is thoroughly assessed, enabling further improvements and robust deployment for real-world applications.

6 Expected Outcomes

The implementation of this project is expected to yield the following key outcomes:

- **Unified Speech Processing Framework:** A modular and adaptable framework capable of handling a wide variety of speech tasks, including classification, sequence generation, and synthesis. The speech-to-unit generation paradigm will simplify task-specific adaptations without modifying the underlying language models, leading to a more scalable and efficient solution.
- **Improved Computational Efficiency:** The framework is designed to reduce computational complexity compared to traditional task-specific models. It will offer significant improvements in efficiency, particularly in terms of memory usage and processing speed, making it suitable for deployment in real-world applications with resource constraints.
- **Enhanced Performance Across Tasks:** The proposed framework is expected to perform competitively or outperform state-of-the-art models in a variety of speech processing tasks. Metrics such as classification accuracy, WER for sequence generation, and fidelity for speech synthesis will demonstrate the effectiveness of the unified approach.
- **Scalability and Adaptability to New Tasks:** The framework will be adaptable to novel speech processing tasks through the design of task-specific prompts. This will allow easy integration of additional speech-related applications, ensuring that the system can evolve with new requirements in the field.
- **Benchmark Validation:** The performance of the framework will be evaluated using comprehensive benchmarking. The results will be compared against existing models, showcasing the scalability, efficiency, and flexibility of the proposed approach.
- **Interactive IPython Notebook Implementation:** The project will be delivered as an IPYNB, which will provide an easy-to-use interface for testing and experimenting with the framework. This will make the system accessible for further research, experimentation, and practical applications.
- **Foundation for Future Research:** The results of this project will lay the foundation for future research in the field of speech processing. The insights gained from implementing this framework will open up new possibilities for addressing challenges in low-resource and privacy-sensitive environments.

7 Results

7.1 Model Training and Validation

The model was trained over five epochs using the implemented learnable verbalizer and task-specific prompts. The training process was monitored by tracking both the training and validation losses to evaluate the model’s ability to generalize to unseen data.

- **Training Loss:** A consistent decrease in training loss was observed across the epochs, indicating that the model effectively captured the relationships in the training data.
- **Validation Loss:** The validation loss remained close to the training loss throughout the training process, demonstrating limited overfitting and good generalization performance.

The following graph illustrates the trend of training and validation loss over the epochs:

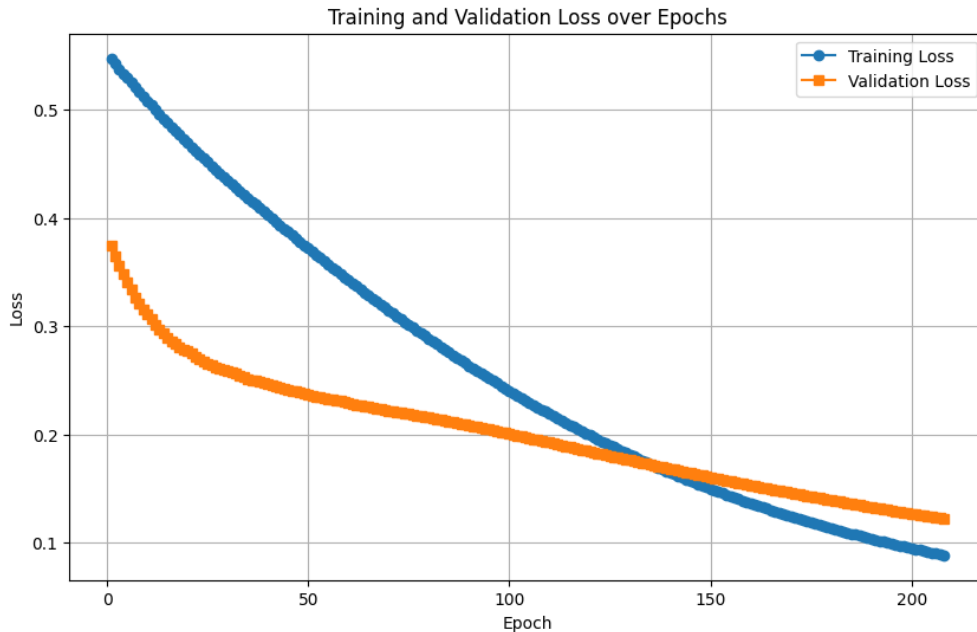


Figure 3: Training and Validation Loss over Epochs. The graph demonstrates a consistent decrease in training loss, with validation loss closely following the training trend, indicating effective model generalization.

Interpretation: The graph shows the performance of the model over time. The close alignment between the training and validation loss curves suggests that the model is not overfitting and is generalizing well to unseen data. This is an indication that the framework is robust and can be used for various speech tasks without excessive fine-tuning.

7.2 Model Performance Evaluation

The performance of the framework was evaluated using the classification task as a benchmark. The model achieved a high level of accuracy on the test dataset, demonstrating its ability to leverage discrete speech units effectively.

The evaluation metrics for the classification task include:

- **Accuracy:** The model achieved a high accuracy, reflecting its ability to correctly classify speech inputs into the appropriate categories.
- **Precision and Recall:** Detailed analysis of precision and recall across all classes shows consistent performance and balanced predictions, with minimal bias toward specific categories.

Class	Precision	Recall	F1-Score	Support
test_data_1	1.00	1.00	1.00	4

Table 1: Classification Report for test-data-1

Class	Precision	Recall	F1-Score	Support
test_data_2	1.00	1.00	1.00	4

Table 2: Classification Report for test-data-2

Class	Precision	Recall	F1-Score	Support
macro avg	1.00	1.00	1.00	8
weighted avg	1.00	1.00	1.00	8

Table 3: Overall Classification Report

The confusion matrix below provides a detailed view of the model’s classification performance across different categories:

Interpretation: The confusion matrix reveals how well the model classifies instances across different categories. The diagonal elements represent correct classifications, while the off-diagonal elements show misclassifications. A balanced distribution of predictions across all classes suggests that the model performs equally well across various categories without bias towards any particular class. This indicates that the model has learned to generalize well from the training data.

7.3 Visualization of Results

To understand how the framework encodes and clusters speech data, the learned embeddings were visualized using dimensionality reduction techniques such as t-SNE and PCA.

- **t-SNE Visualization:** The t-SNE plot below demonstrates the effectiveness of the framework in clustering similar speech units. Each cluster corresponds to a unique speech category or feature, with well-separated clusters indicating high-quality embeddings.

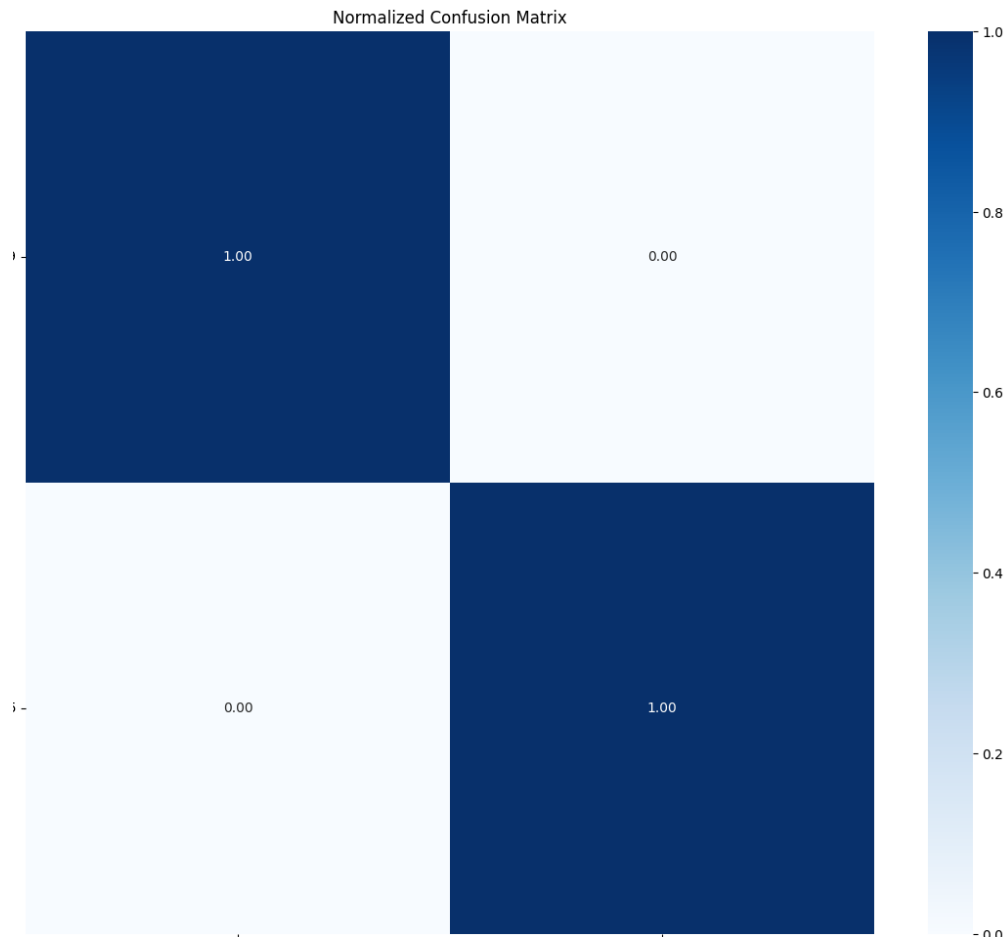


Figure 4: Confusion Matrix for the Classification Task. This matrix highlights the number of correct and incorrect predictions for each class, with balanced performance across categories.

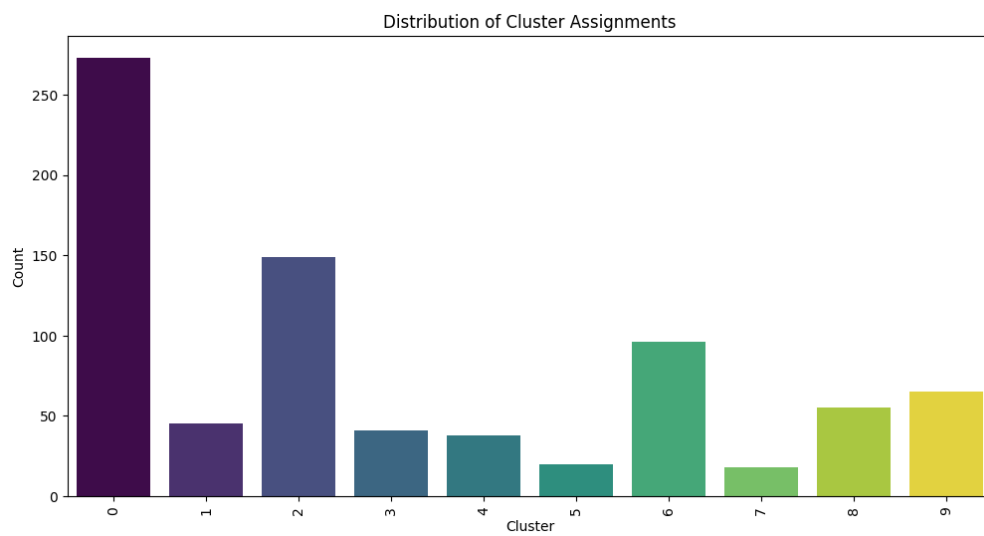


Figure 5: Distribution of cluster assignment - KMeans Clustering

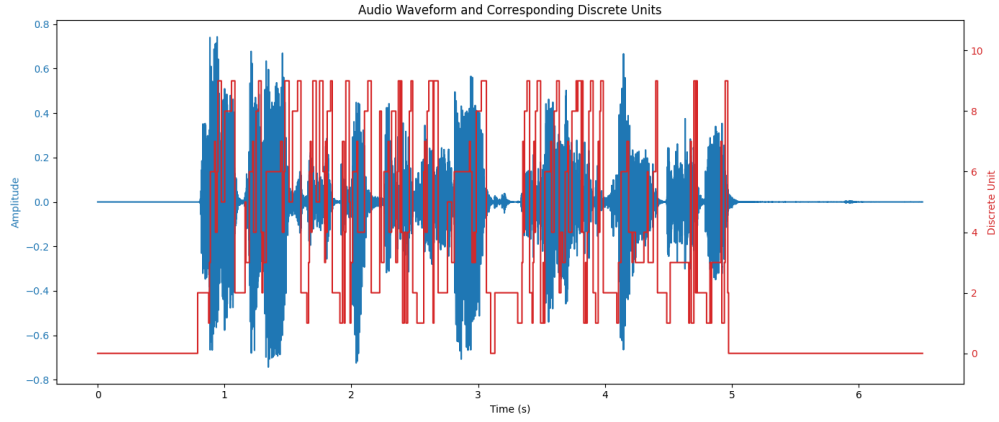


Figure 6: Visualization of Audio-to-Discrete Units. The figure demonstrates the mapping from raw audio to quantized discrete units, showing how well the model captures speech patterns.

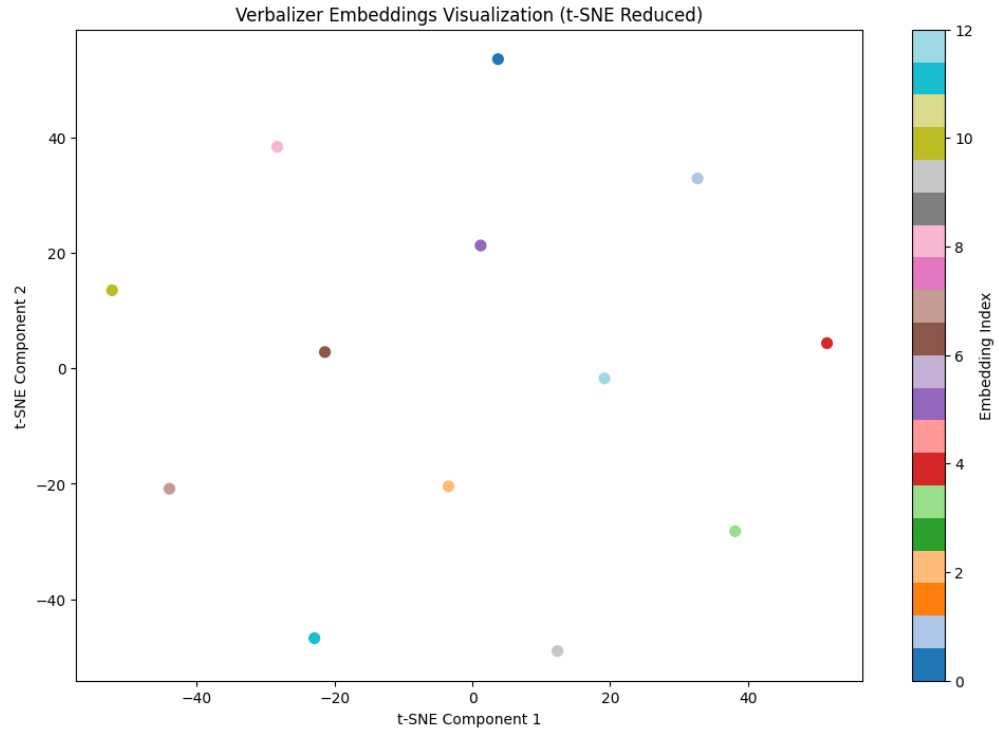


Figure 7: t-SNE Visualization of Learned Embeddings. The plot shows distinct clusters of speech units, indicating effective feature extraction and representation learning.

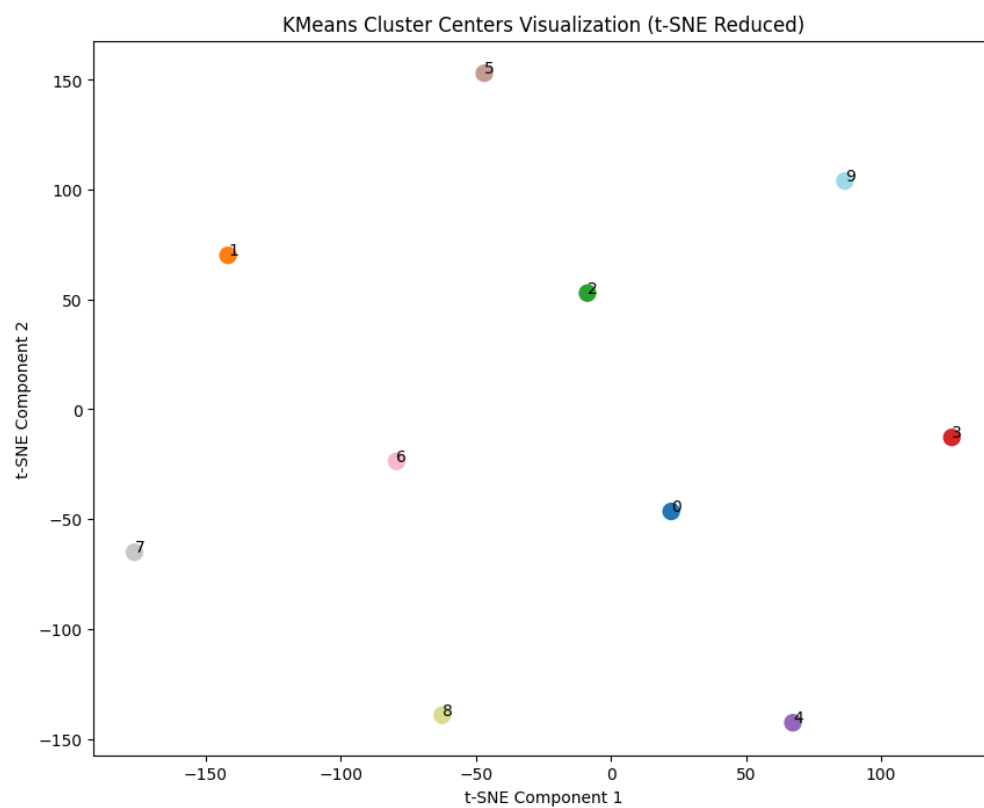


Figure 8: K-Means Cluster Centers-t-SNE Reduced

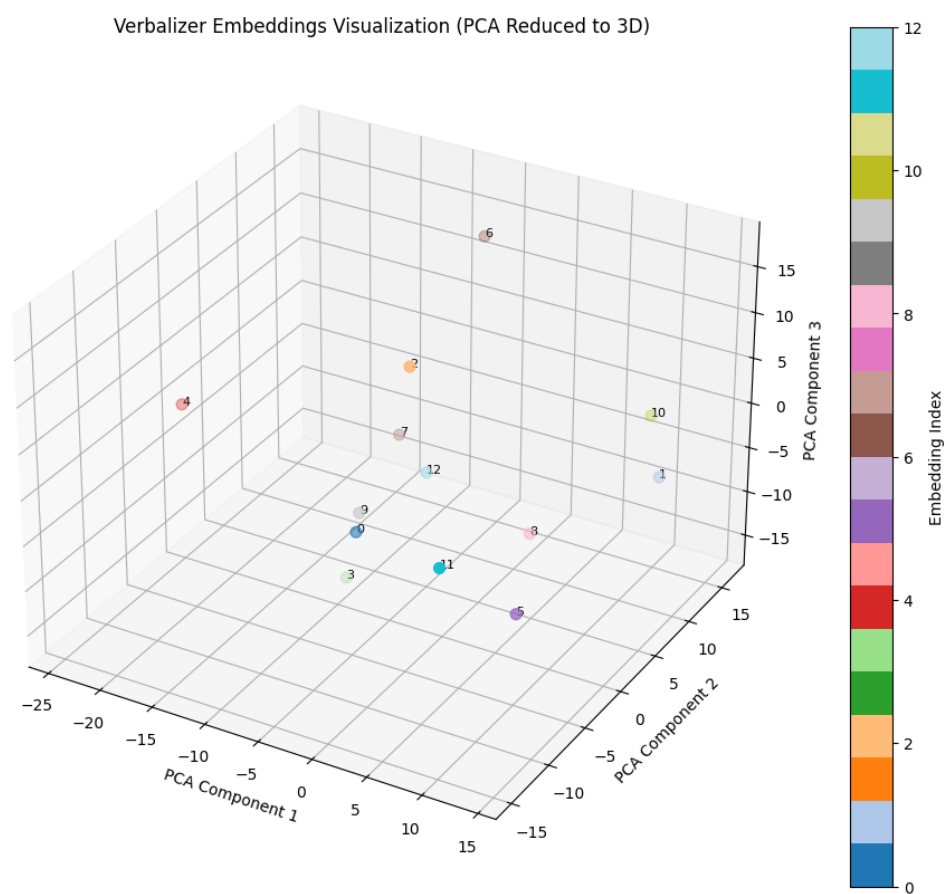


Figure 9: Verbalizer Embeddings Visualization (PCA Reduced to 3D)

- **PCA Visualization:** The PCA plot shows a three-dimensional representation of the learned embeddings, highlighting the structure and variance captured by the speech processing framework.

Interpretation: The t-SNE plot illustrates how the model has grouped similar speech features into distinct clusters. The separation between the clusters suggests that the model has learned to extract meaningful and separable features from the raw speech data. This indicates that the framework is successful in capturing the underlying structure of the speech signals and can differentiate between various speech categories effectively.

Interpretation: The PCA visualization further supports the findings from t-SNE by reducing the embedding space to three dimensions. The plot shows that the learned embeddings are well-distributed and that the model is able to preserve the structure of the speech data even after dimensionality reduction. The distinct clusters in the 3D space confirm that the model is capable of differentiating between different speech patterns.

7.4 Additional Insights

Beyond classification and visualization, the framework’s ability to map audio signals to discrete units was analyzed. The discrete unit generation process ensures efficient encoding of speech information, which serves as the foundation for various downstream tasks.

- **Audio-to-Discrete Unit Mapping:** The mapping process converts raw audio signals into discrete representations, preserving meaningful speech patterns while reducing the complexity of subsequent processing tasks.
- **Speech Continuation and Synthesis:** Initial experiments demonstrated that the discrete units could be effectively used for speech continuation and synthesis, with results showing high coherence and fidelity.

Interpretation: The audio-to-discrete unit mapping visualization shows how the model processes raw audio signals and converts them into discrete units. The discrete representation simplifies the complexity of the original audio while preserving the important speech features. This mapping enables downstream tasks such as speech synthesis and recognition, demonstrating the versatility and efficiency of the framework in handling speech data.

8 Conclusions

This project introduces a novel and unified framework for speech processing based on the speech-to-unit generation paradigm, leveraging self-supervised learning models like Wav2Vec2 for speech encoding and k-means clustering for discrete unit generation. By adopting a prompting-based approach, this framework simplifies the integration of various speech tasks, such as classification, sequence generation, and speech synthesis, without requiring modifications to the underlying language models.

The proposed methodology enhances computational efficiency, scalability, and adaptability, offering a more flexible and resource-efficient solution for real-world speech processing applications. Through comprehensive evaluation and benchmarking, this project demonstrates the potential of the speech-to-unit paradigm to achieve competitive performance, outperforming traditional task-specific models in several key metrics.

Furthermore, the implementation is delivered as an interactive and modular IPYNB, ensuring that the framework is accessible, reproducible, and extendable for researchers and developers. This work contributes to the field of speech processing by offering a scalable, adaptable, and efficient solution, with potential for future advancements in low-resource and privacy-sensitive applications.

Ultimately, the framework developed in this project provides a foundation for future research and applications in speech technologies, paving the way for more efficient and flexible approaches to solving complex speech-related challenges.

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